

Prajit Rajkumar

510-738-6009 | prajkumar@ucsd.edu | prajit.rjk@gmail.com | [linkedin.com/in/prajitrajkumar](https://www.linkedin.com/in/prajitrajkumar) | github.com/prajitrr
Website: prajit.it | Google Scholar: [prajit.it/scholar](https://scholar.google.com/citations?user=prajit)

EDUCATION

University of California, San Diego [UCSD] (GPA: 3.83, Provost Honors)

B.S. Mathematics-Computer Science

La Jolla, CA

Sep. 2023– Jun. 2027

B.S. Biology with a Specialization in Bioinformatics

Sep. 2023– Jun. 2027

Mission San Jose High School [MSJHS] (GPA: 3.97)

Fremont, CA

EXPERIENCE

Enveda

Jun. 2025 – Sep. 2025

Data Science Intern

Boulder, CO

- Led benchmarking study for first stage of Enveda's computational drug discovery pipeline; planned high-impact publication
- Optimized internally used software by 60x, accelerating multiple concurrent projects
- Developed novel metrics including improved clustering method for quantifying drug candidate search space

Dorrestein Lab

Aug. 2023 – Present

Undergraduate Researcher

La Jolla, CA

- Computational mass spectrometry/metabolomics projects w/ applications to clinical medicine and drug discovery
- SWE & data science for multiple high-impact publications in top tier journals including *Cell* and *Nature Protocols*
- Awarded grants from *Google Deepmind* and *OpenAI*; submitted independent research to *Analytical Chemistry*; invited to present at Metabolomics Society conference

King Lab

Jan. 2025 – June 2025

Undergraduate Researcher

La Jolla, CA

- Applied ML/computer vision with vision transformers on whole-slide histology images to understand prognosis of heart failure
- Introduced 7x speedup to large-scale image preprocessing workflow, increasing data pipeline throughput

UCSD Academic Achievement Hub

Jun. 2024 – Present

Supplemental Instruction Leader

La Jolla, CA

- Lead supplemental instruction sessions for various engineering calculus courses, facilitating student learning and problem-solving

PUBLICATIONS

Rajkumar, P., Tang, R., Sapre, H., Zemlin, J., Deleray, V., Seo, J.I., Mohan, S., Xing, S., Gouda, H., Abiead, Y.E., Tsunoda, S.M., Zhao, H.N., *et al.* (2026). Retrieval-Augmented Language Models Enable Scalable Chemical Source Classification in Metabolomics Workflows. **Analytical Chemistry**. <https://doi.org/10.1021/acs.analchem.5c05301>. {Accepted, Pending release on website}

Mannochio-Russo, H., Charron-Lamoureux, V., van Faassen, M., Lamichhane, S., Gonçalves Nunes, W. D., Deleray, V., Ayala, A. V., Tanaka, Y., Patan, A., Vittali, K., **Rajkumar, P.**, *et al.* (2025). The microbiome diversifies long- to short-chain fatty acid-derived N-acyl lipids. **Cell**, 188(15), 4154–4169.e19. <https://doi.org/10.1016/j.cell.2025.05.015>.

Charron-Lamoureux, V., Mannochio-Russo, H., Lamichhane, S., Xing, S., Patan, A., Portal Gomes, P. W., **Rajkumar, P.**, *et al.* (2025). A guide to reverse metabolomics—a framework for big data discovery strategy. **Nature Protocols**. <https://doi.org/10.1038/s41596-024-01136-2>.

Zhao, H. N., *et al.* (2025). A resource to empirically establish drug exposure records directly from untargeted metabolomics data. **Nature Communications**. <https://doi.org/10.1038/s41467-025-65993-5>.

Krutkin, D. D., Thomas, S., Zuffa, S., **Rajkumar, P.**, *et al.* (2025). To Impute or Not To Impute in Untargeted Metabolomics-That is the Compositional Question. **Journal of the American Society for Mass Spectrometry**. <https://doi.org/10.1021/jasms.4c00434>.

Patan, A., Xing, S., Charron-Lamoureux, V., Hu, Z., Deleray, V., Agongo, J., El Abiead, Y., Mannochio-Russo, H., Mohanty, I., Gouda, H., Zemlin, J., **Rajkumar, P.**, *et al.* (2025). Charting the undiscovered metabolome with synthetic multiplexing. *BioRxiv*. <http://doi.org/10.1101/2025.11.18.689170>.

Mannochio-Russo, H., Gonçalves Nunes, W.D., Zhao, H.N., Kvitne, K.E., Xing, S., Gouda, H., Agongo, J., Mohanty, I., Charron-Lamoureux, V., **Rajkumar, P.**, *et al.* (2025). Bridging Complexity and Accessibility in Metabolomics with MetaboApps. *ChemRxiv*. <https://doi.org/10.26434/chemrxiv-2025-3nq29>.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, Java, R, SystemVerilog

Dev. Tools: Git, Bash, Docker, Google Cloud Platform, Amazon Web Services (AWS), Microsoft Azure, Databricks, VS Code

Libraries/Misc: PyTorch, pandas, NumPy, scikit-learn, scipy, matplotlib, seaborn, HuggingFace, PEFT / LaTeX, Onshape, Arduino

AWARDS

ML: YCombinator AI Startup School (YC) [\$25,000 Grant], Google Deepmind Gemma Grant (\$10,000), OpenAI Researcher Grant

Academic: Goldwater Scholarship Campus Nominee, USA Biology Olympiad Semifinals Honorable Mention (Top 125 USA), American Invitational Mathematics Examination (AIME) Qualifier

SCHOOL

Selected Coursework: –*Graduate*– ECE 284 (Low-Power VLSI for Machine Learning), CSE 284 (Genomics for Bioinf.), BENG 294A (Patient-Centered Clinical Medicine), BENG 294B (Engineering in the Patient Healthcare system), BENG 230C (Quantitative Cardiovascular Physiology), PHYS 141/241 (Computational Physics) –*Undergrad*– CSE 100/101 (Advanced Data Structures/Algorithms)

Selected Community: *Vice President, Finance* - Undergraduate Bioinformatics Club (UBIC) @ UCSD; UCSD Peer Mentor